

to withstand extremely hostile temperatures that vary from super-cooled temperatures before ignition to super-hot ones when it fires, metallurgy was a key hurdle that had to be overcome. The right alloys had to be developed that could take these massive temperature fluctuations and also keep the weight of the engine down.

There were other complications. The problem with liquid hydrogen is that it has a tendency to return to its gaseous form rapidly and is highly inflammable. It ignites spontaneously when it comes in contact with atmosphere. So the temperature has to be

the Russians to supply five more cryogenic engines to power its GSLV flights at a cost of \$5 million (Rs 20 crore) each.

It was only in 2006 that the ISRO team felt it had, to an extent, mastered the range of technologies needed for the test. The first firing of the engine in October 2006 went off smoothly for the planned 50-second duration. Emboldened, the scientists tried for a longer duration in January 2007 but had to abort it within 30 seconds as an extraneous fire occurred in one of the pipelines. More recently in August this year, the system fired smoothly for 500 seconds but because of a technical

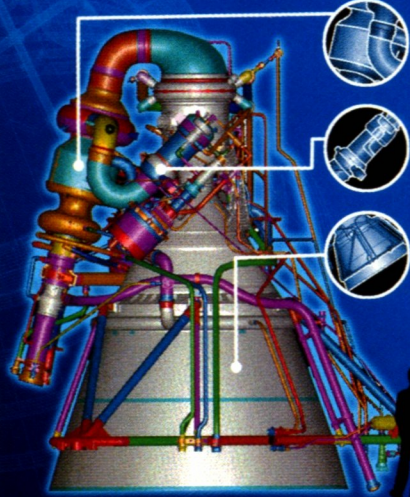
from the international market is proving increasingly difficult. He says, "Not only were other nations asking for the moon in terms of price but they were also not available for the next five years." He denies that India took Russian help to develop the engine as alleged by some scientists and points out that ISRO not only indigenously developed a host of new materials needed for the engine but also the various critical components for it.

Much more though needs to be done. The research organisation is in the process of building the more powerful GSLV Mark III launch vehicle capa-

Graphic by RAM

CRYO BREAKTHROUGH

ISRO took 13 years to master the range of technologies involved in such engines



TURBO PUMPS: They operate at very high RPMs.

PIPING: Joints have to be specially welded.

CASING: Made from special alloys that proved to be a major challenge.

ENGINE: It has to withstand both extreme hot and cold temperatures.

WHAT ARE CRYOGENIC ENGINES?

By using supercooled liquid hydrogen and oxygen as propellants, they give more bounce per ounce of fuel used as compared to solid or liquid fuel rockets.



PROPELLANTS: The liquid hydrogen has to be stored at minus 250°C. It is highly inflammable and has to be handled with great care.



THE SAVINGS: By making such engines indigenously the country saves foreign exchange and also avoids the long delays in having them imported.

tightly monitored and the pipelines and pumps through which liquid hydrogen is fed have to be totally leak proof. This meant that joints had to be welded rather than screwed tight. The cryogenic engine's project director Mohammed Muslim says, "There is absolutely no margin for error." Other major hurdles were developing the thrust chamber, which required intricate grooves to be machined, and the high speed turbo pumps.

Initial efforts by ISRO engineers met with little success. The first test that took place in 1999 ended in abject failure when one of the gas generators blew on the test stand. With delays inevitable, in 2001 ISRO negotiated with

glitch, it had to be aborted before completion. After identifying the problem, the scientists felt that it was a minor issue and so scheduled another test for mid-November. They waited anxiously at the Mahendragiri test stand to watch as the countdown began for the engine. This time the test was flawless.

ISRO Chairman Madhavan Nair, who was there too, told INDIA TODAY: "For ISRO, it is the culmination of the development of rocket technology—we are now totally self-reliant in the entire range of rocketry whether solid, liquid or cryogenic." Nair points out that the development was critical not just because of the foreign exchange it saves but also because getting cryogenic engines

ble of putting a four-tonne INSAT class satellite in orbit, almost twice the weight of its current capacity. For this, an all new heavy cryogenic engine is being developed. The first test is being planned next year. Even the recently tested cryogenic engine has to be modified if it has to fly for a future mission to carry slightly heavier satellites than the existing two-tonne class.

More importantly, the proof of ISRO's capability would be demonstrated only when the cryogenic engine that it has developed flies on the GSLV and performs perfectly. Till then, ISRO scientists can take the much deserved credit while keeping their fingers firmly crossed. ■